

Gene Therapy for XLRP *RPGR*

An open label, multi-centre, Phase I/II dose escalation trial of a recombinant adeno-associated virus vector (AAV2/5-hRKp.RPGR) for gene therapy of adults and children with X-linked Retinitis Pigmentosa owing to defects in Retinitis Pigmentosa GTPase Regulator (*RPGR*)

Version	11.0
Date	1 st May 2020
Sponsor	MeiraGTx UK II Ltd
Sponsor registration #	MGT009
Trial registration	EudraCT 2016 003967 21
CTA #	45522/0004/001 0001
NRES #	218294

Authorisation: Co-ordinating Investigator

Name
Role

Signature

Date

Authorisation: Sponsor Representative

Name
Role

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1.3 Structured trial summary

Primary Registry and Trial Identifying Number	EudraCT 2016-003967-21
Secondary Identifying Numbers	MeiraGTx UK II Ltd. registration number: MGT009
Sponsor	MeiraGTx UK II Ltd.
Contact for Public Queries	ocularinfo@meiragtx.com
Contact for Scientific Queries	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
Public Title	<i>RPGR</i> XLRP trial
Scientific Title	An open label, multi-centre, Phase I/II dose escalation trial of a recombinant adeno-associated virus vector AAV2/5-hRkp.RPGR for gene therapy of adults and children with X-linked Retinitis Pigmentosa owing to defects in <i>RPGR</i>
Countries of Recruitment	United Kingdom, United States of America
Health Condition(s) or Problem(s) Studied	X-linked Retinitis Pigmentosa
Intervention(s)	Open label, dose-escalation • [REDACTED] [REDACTED] [REDACTED]
Key Inclusion and Exclusion Criteria	Key Inclusion Criteria: • Males aged 5 years or older (children will be included only once the maximal tolerated dose has been determined) • With Retinitis Pigmentosa caused by mutations in <i>RPGR</i> • Evidence of relative preservation of retinal structure at the macula • Able to undertake age-appropriate clinical assessments • Willing to give consent for the use of blood and blood components collected throughout the trial for the investigation of immune response to ATIMP Key Exclusion Criteria: • Intra-ocular surgery within 3 months of screening • Ocular or systemic disorder that may preclude subretinal surgery and/or interfere with interpretation of the study results. • Participated in another research study involving an investigational therapy for ocular disease within the last 6 months

	- Have any other condition that the Principal Investigator (PI) considers makes them inappropriate for entry into the trial - Are unwilling to consider the possibility of entry into a subsequent longer term follow up study
Study Type	Phase I/II, open-label, multi-centre, dose escalation in adults, followed by dose confirmation in children, followed by further randomised dose confirmation against a control arm in adults and children with XLRP owing to defects in <i>RPGR</i>
Date of First Enrolment	Q3 2017
Target Sample Size	Up to 71 participants
Primary Outcome(s)	The primary outcome is safety of subretinal administration of AAV2/5-hRKp.RPGR. Safety is defined as the absence of ATIMP-related: - Reduction in visual acuity by 15 ETDRS letters or more - Severe unresponsive inflammation - Infective endophthalmitis - Ocular malignancy - Grade III or above non-ocular SUSAR Safety will be assessed for 12 months after the intervention in this study, and a further 4 years in a separate follow on study (for deferred arm patients' safety will be assessed for 6 months after the intervention in this study and a further 4½ years in a separate follow on study).
Key Secondary Outcomes	The secondary outcomes are measures of the efficacy of the intervention, which will be performed on an individual participant basis and will be descriptive in nature. Efficacy will be assessed at several time points between 3 to 12 months after the intervention: 1) Slowing or halting of progressive deterioration in retinal structure or visual function that is greater than the baseline variation for that test and is sustained for at least two consecutive assessments. Slowing or halting of progression over time may be facilitated by comparing identical structural and functional assessments acquired prior to intervention/injection to better determine rate of change over time in this slowly progressive disease. 2) Any improvement in visual function from baseline that is greater than the baseline variation for that test and is sustained for at least two consecutive assessments. 3) Any improvement in retinal function from pre-intervention that is greater than baseline variation and measurable by electrophysiology (pattern ERG, multifocal ERG or full-field ERG). 4) Health- and vision-related quality of life measures (HRQoL, VRQoL).

1.4 Roles and responsibilities

Agreements that include detailed roles and responsibilities will be in place between participating sites and MeiraGTx UK II Ltd.

These membership lists are correct at the time of writing; please see terms of reference documentation in the TMF for current lists.

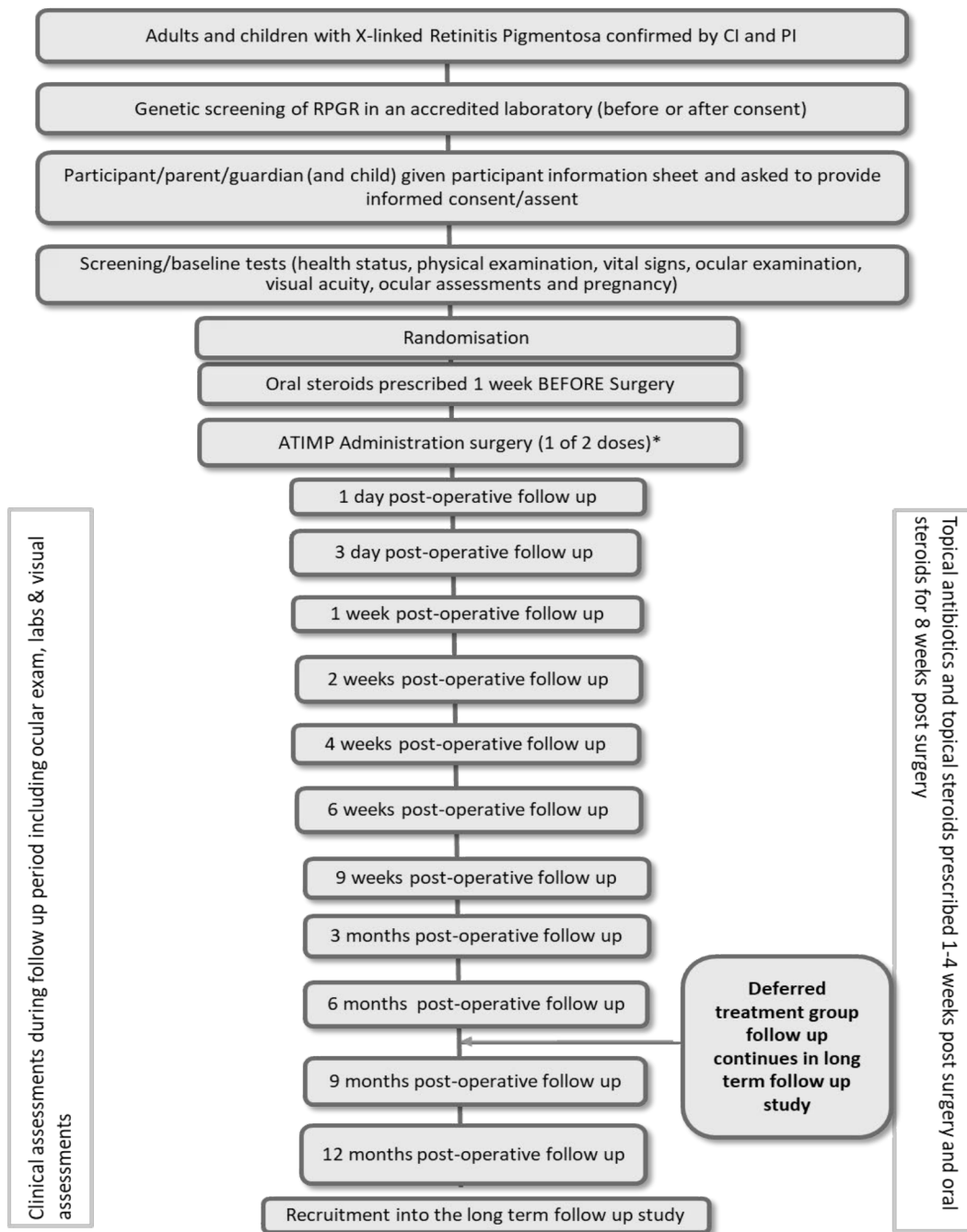
1.4.1 Protocol contributors

Name	Affiliation	Role
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]

1.4.2 Independent Data Monitoring Committee

Name	Affiliation	Role and responsibilities
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]
[REDACTED]	[REDACTED]	[REDACTED]

2 Trial Diagram



* If randomised to the deferred treatment arm ATIMP administration will occur 6 months after screening, follow-up post administration will be for 6 months

3 Abbreviations

AAV	Adeno-Associated Virus
Ad	Adenovirus
AE	Adverse Event
AF	Autofluorescence
AO	Adaptive Optics
AR	Adverse Reaction
ATIMP	Advanced Therapy Investigational Medicinal Product
bps	Basepairs
BGL65p	<i>hRPE65</i> promoter fragment that is less efficient than the NA65p promoter
BNF	British National Formulary
CAI	Codon Adaptation Index
cDNA	complementary Deoxyribonucleic Acid
CGIC	Clinician Global Impression of Change
CI	Chief Investigator
CMT	Clinical Management Team
CMV	Cytomegalovirus
CNS	Central nervous system
CRF	Case Report Form
CRO	Contract Research Organization
CTA	Clinical Trial Authorisation
CTIMP	Clinical Trial of Investigational Medicinal Product
DAPI	4',6-diamidino-2-phenylindole
DLE	Dose-limiting event
DNA	Deoxyribonucleic acid
DSUR	Development Safety Update Report
EC	European Commission
eGFP	enhanced green fluorescent protein
ELISA	Enzyme-linked Immunosorbent Assay
ELISPOT	Enzyme-linked ImmunoSpot Assay
EMA	European Medicines Agency
EOSRDs	early-onset severe retinal dystrophies

EQ-5D-5L	EuroQOL Quality of Life 5 dimension 5 level
ERG	Electroretinography
ETDRS	Early Treatment Diabetic Retinopathy Study
EU	European Union
EUCTD	European Clinical Trials Directive
EudraCT	European Clinical Trials Database
EudraVIGILANCE	European database for Pharmacovigilance
FAF	Fundus Autofluorescence
FDA	(US) Food and Drug Administration
FWA	Federal Wide Assurance
GCP	Good Clinical Practice
GFP	Green Fluorescent Protein
GLP	Good Laboratory Practice
GMO	Genetically Modified Organism
GMP	Good Manufacturing Practice
GMSC	Genetic Modification Safety Committee
GTAC	Gene Therapy Advisory Committee
H&E	haematoxylin and eosin
hEBs	Human embryonic bodies
HEK293T	human embryonic kidney epithelial cell line
<i>hRPE65</i>	human retinal pigment epitheliumspecific protein 65 (all- <i>trans</i> -retinyl isomerase)
HSE	Health and Safety Executive
HTA	Human Tissue Authority
H2B	histone H2B
IB	Investigator Brochure
ICF	Informed Consent Form
ICH	International Conference on Harmonisation
IDMC	Independent Data Monitoring Committee
IMP	Investigational Medicinal Product
IMPD	Investigational Medicinal Product Dossier

IND	Investigational New Drug
IRB	Institutional Review Board
ISF	Investigator Site File
ISRCTN	International Standard Randomised Trial Number
ITRs	Inverted terminal repeats
ITT	Intention to Treat
IV (or iv)	Intravenously
IVI	Impact of Visual Impairment
■	■
LCA	Leber congenital amaurosis
LCA2	Retinal dystrophy associated with defects in <i>RPE65</i>
LLQ	Low Luminance Questionnaire
LREC	Local Research Ethics Committee
MA	Marketing authorisation
Main REC	Main Research Ethics Committee
MHRA	Medicines and Healthcare products Regulatory Agency
MRI	Magnetic Resonance Imaging
mRNA	Messenger RNA
NA65p	Optimised <i>hRPE65</i> promoter
NHS R&D	National Health Service Research & Development
NIH	National Institutes of Health
NIMP	Non-Investigational Medicinal Product
NRES	NHS National Research Ethics Service
NZW	New Zealand White
OBA	Office of Biotechnology Activities
OCT	Optical Coherence Tomography
ONL	Outer nuclear layer
ORF15	open reading frame 15
PCT	Polymerase Chain Reaction
PERG	Pattern Electroretinogram
PGIC	Patient Global Impression of Change
PGIS	Patient Global Impression of Severity
PHI	Protected Health Information
PI	Principal Investigator
PIS	Participant Information Sheet

QA	Quality Assurance
QALY	Quality Adjusted Life Year
QC	Quality Control
qds	quarter die sumendus (four times a day)
QOL	Quality of Life
QMMP	Quality Management and Monitoring Plan
qPCR	Quantitative Polymerase Chain Reaction
QP	Qualified Person for release of CTIMP
qPCR	Quantitative polymerase chain reaction
rAAV	Recombinant adeno associated virus
rAAV2	recombinant adeno-associated virus serotype 2
R&D	Research and Development
REC	Research Ethics Committee
RG	Research grade
RK	Rhodopsin Kinase
RNA	Ribonucleic acid
RPE	Retinal Pigment Epithelium
RPE65	retinal pigment epithelium specific protein 65 (all- <i>trans</i> -retinyl isomerase)
RPGR	retinitis pigmentosa GTPase regulator
rt-PCR	reverse transcription polymerase chain reaction
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SAR	Serious Adverse Reaction
SD	standard deviation
SDV	Source Data Verification
SOP	Standard Operating Procedure
SmPC	Summary of Product Characteristics
SSA	Site Specific Approval
SSAR	Suspected Serious Adverse Reaction
SUSAR	Suspected Unexpected Serious Adverse Reaction
SV40	Simian virus 40
TEI	Treatment Experience Interview

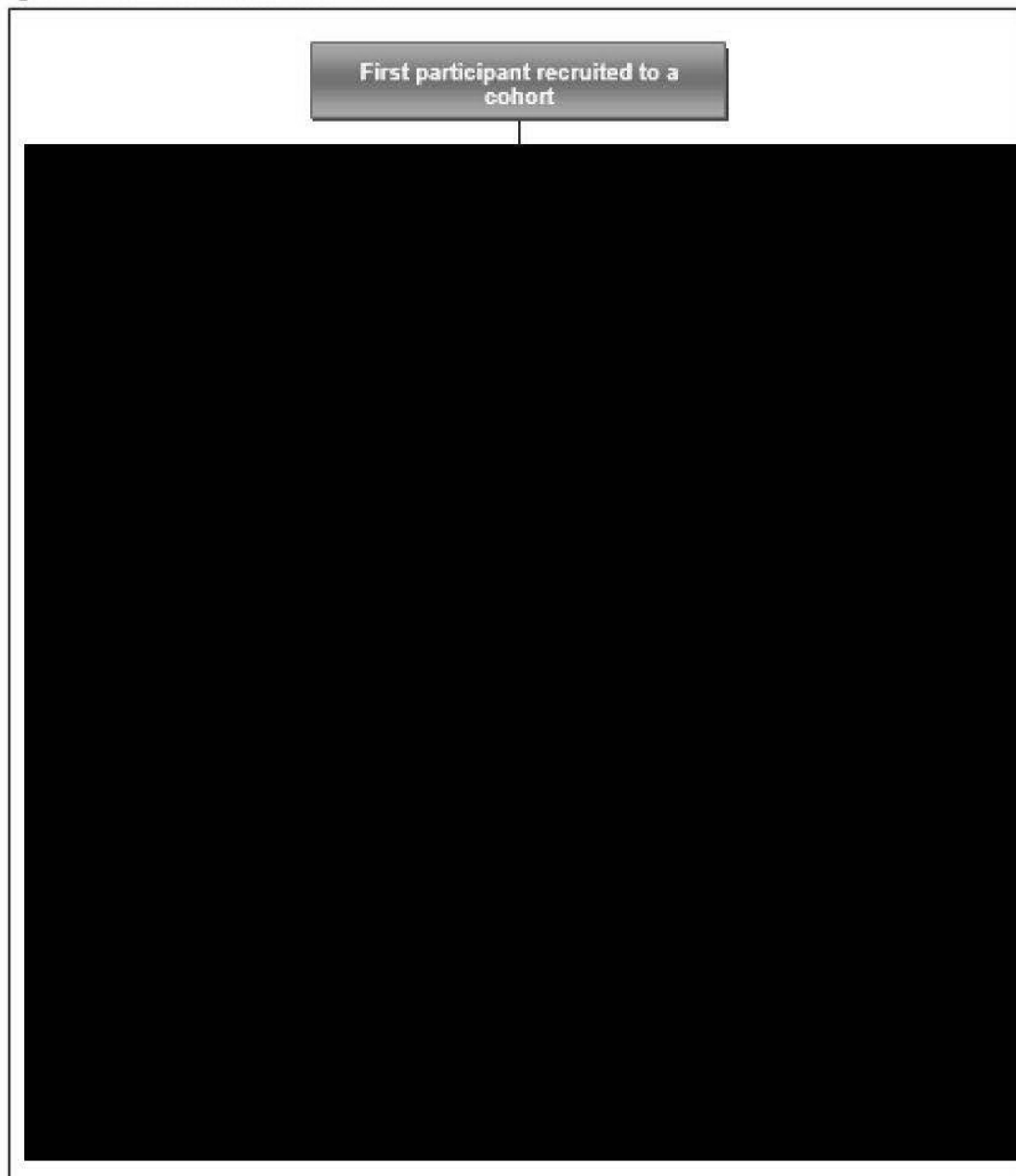
MGT009, Gene Therapy for XLRP *RPGR*

TMF	Trial Master File
ToR	Terms of Reference
tRNA	Transfer ribonucleic acid
UAR	Unexpected Adverse Reaction
█	█
XLRP	X-linked retinitis pigmentosa

USA	United States of America
vg	Viral Genomes
█	█
WT	Wild type

Number of DLEs			
Low dose			
Participants 1-3	Participants 4-6	Action	Details
[REDACTED]		[REDACTED]	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
[REDACTED]		[REDACTED]	[REDACTED]
[REDACTED]		[REDACTED]	[REDACTED]
[REDACTED]		[REDACTED]	[REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED] [REDACTED] [REDACTED] [REDACTED] [REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED]
Intermediate dose			
Participants 1-3	Participants 4-6	Action	Details
[REDACTED]		[REDACTED]	[REDACTED]
[REDACTED]		[REDACTED]	[REDACTED]
[REDACTED]		[REDACTED]	[REDACTED] [REDACTED] [REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED] [REDACTED] [REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED] [REDACTED]
High dose			
Participants 1-3	Participants >3	Action	Details
[REDACTED]		[REDACTED]	[REDACTED] [REDACTED]
[REDACTED]		[REDACTED]	[REDACTED] [REDACTED]
[REDACTED]		[REDACTED]	[REDACTED] [REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED]
	[REDACTED]	[REDACTED]	[REDACTED] [REDACTED] [REDACTED] [REDACTED]

Figure 1: Dose escalation Flow chart



4.3.2.2.5 Confirmatory Safety Phase

The adult dose escalation component of the study is complete at the time of protocol amendment v9.0. Review of the data collected to date with the IDMC led to a decision not to pursue the higher dose level and to commence administration for confirmation in paediatrics at the intermediate dose level. Children are defined as those aged 15 and under in the UK, and 17 and under in the US. These participants in the paediatric confirmation phase are administered the intermediate dose [REDACTED] [REDACTED] [REDACTED] which has

Figure 2: Randomisation Timeline

Time (months)	
0	Participant completes screening Randomisation 1: Eye selected for administration Randomisation 2: Treatment Group • Low • Intermediate • Deferred (Control Group) Randomisation 3: Dose selected for administration to Deferred (Control Group) • Low • Intermediate
6 months after completion of baseline assessments in deferred treatment group	Control group administered low or intermediate dose Interim analysis performed 3 and 6 months post- surgery of the last participant in the low or intermediate treatment arm
12	Follow-up complete

5 Methods

5.1 Site Selection

The trial sponsor MeiraGTx UK II Ltd. has overall responsibility for site and investigator selection.

5.1.1 Study Setting

The study settings are academic hospitals and academic research centres selected for their ability to perform the intervention and assessments required of this protocol. Data is anticipated to be collected from centres in the United Kingdom and the United States of America.

5.1.2 Site/Investigator Eligibility Criteria

Once a site has been assessed as being suitable to participate in the trial, they will be provided with a copy of this protocol and the ATIMP Investigator Brochure.

To participate in the *RPGR* XLRP trial, investigators and trial sites must fulfil a set of criteria that have been agreed by the Clinical Management Team (CMT) and that are defined below.

Eligibility criteria:

- A named clinician is willing and appropriate to take Principal Investigator responsibility
- Suitably trained staff are available to recruit participants, enter data and collect samples
- Suitably trained and certified staff are available to undertake clinical assessments

Terminal Repeats) sequences that provide *in cis* the packaging signal, a cDNA encoding human *RPGR-ORF15*, driven by a human photoreceptor-specific genomic promoter (Rhodopsin Kinase, RK, *GRK1*). The icosahedral capsid consists of three related capsid proteins, VP1, VP2, and VP3. AAV has a compact macromolecular structure and forms stable viral particles 20nm in diameter. The vector particles are replication incompetent.

5.4.1.2 ATIMPs Classified as Genetically Modified Organisms

The ATIMP (AAV2/5-hRKp.RPGR) is classified as a genetically modified organism under the Genetically Modified Organisms (Contained Use) Regulations 2000.

The Health and Safety Executive (HSE) must be notified of each UK clinical trial site administering the ATIMP for first use of premises for genetic modification activities before the activities commence. A risk assessment of the activities has been carried out and has been reviewed by the local Genetic Modification Safety Committee (GMSC). Internal approval at site for the GMO activities has been gained.

Each clinical trial site administering the ATIMP in the USA must obtain Institutional Biosafety Committee approval in addition to Institutional Review Board (IRB) approval to administer recombinant nucleic acid molecule material to human participants.

5.4.1.3 Source of ATIMPs

The ATIMP has been manufactured at the [REDACTED] in accordance with current Good Manufacturing Practice for clinical trial materials.

5.4.1.4 Preparation and labelling of the ATIMP

Preparation and labelling of the ATIMP will be completed in accordance with the relevant GMP and national guidelines.

The ATIMP is a recombinant serotype 2/5 adeno-associated viral vector containing a human RPGR-ORF15 cDNA, with a 378 bp (in-frame) deletion in the purine-rich tract in the ORF15 region. The transgene is driven by a fragment of the human Rhodopsin Kinase promoter. The manufacturer [REDACTED] will perform analytical testing and provide certification. Aliquoting into vials and labelling of the ATIMP was the

Whole blood will be processed at the [REDACTED], or a CLIA-accredited molecular diagnostic laboratory in the US, for *RPGR* mutation screening. Haematology and biochemistry samples and screen will be carried out at the [REDACTED], or an accredited laboratory in the US (Table 2 Haematology and Biochemistry).

Table 2: Haematology and Biochemistry

<u>Haematology</u>	<u>Biochemistry</u>
Haemoglobin	Sodium
Haematocrit	Potassium
Erythrocytes	Chloride
Mean Corpuscular Volume	Bilirubin
Mean Corpuscular Haemoglobin	Alkaline Phosphatase
Platelet	Aspartate Aminotransferase
Leukocytes	Alanine Aminotransferase
Absolute Neutrophils	Lactate Dehydrogenase
Absolute Lymphocytes	Gamma Glutamyl Transferase
Absolute Neutrophils	Protein
Absolute Eosinophils	Albumin
Absolute Basophils	Calcium
	Magnesium
	Phosphate
	Urea
	Urate
	Creatinine
	Glucose

5.4.3 Dispensing

5.4.3.1 Receipt and storage of the Investigational Medicinal Product

[REDACTED]

[REDACTED]

[REDACTED].

The ATIMP vector will be dispensed on the same day as administration according to study specific working instructions, with a 1-hour window for administration after ATIMP has thawed at room temperature.

5.4.4 Dosage and dosage modifications

Trial participants will receive 1 of 3 different doses of ATIMP within the range proven to be safe in the preclinical animal studies.

5.6 Trial Assessments

Table 3: Trial assessments (Dose Escalation Phase, Paediatric Dose Confirmation Phase, and Immediate Treatment Arms of Dose Expansion Phase)*

	Screening	Randomisation	Baseline ^{1, 2}			ATIMP admin	D1	D3	W1	W2	W4	W6	W9	M3	M6	M9	M12
Flexibility of schedule ($\pm$ days)	Up to 4 months prior to randomisation		- 3 months			Day 0	$\pm$ 0D	$\pm$ 1D	$\pm$ 2D	$\pm$ 4D	$\pm$ 7D	$\pm$ 7D	$\pm$ 7D	$\pm$ 14D	$\pm$ 14D	$\pm$ 14D	$\pm$ 14D
Visit number	1		2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Informed consent	•																
Physical exam	•																
Medical history	•								•	•	•	•	•	•	•	•	•
Eligibility determination	•																
IMP administration						★											
Adverse event review			•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Concomitant medication review	•		•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Genetic screening	•																
Vital signs (including height and weight) ³	•		•				•	•	•	•	•	•	•				
Randomisation		★															
Haematology			•				•		•	•	•	•					
Biochemistry/glucose			•				•		•	•	•	•	• ⁷				
Hepatitis B (HBV) testing ⁶	•												•				
Serology			•								•			•	•		
PCR							•				•						
QoL questionnaires (IVI and EQ5D-5L and EQ5D-Y, Low Luminance Questionnaire) Patient Global Impression of Severity			•											•	•		•

	Screening	Randomisation	Baseline ^{1, 2}			ATIMP admin	D1	D3	W1	W2	W4	W6	W9	M3	M6	M9	M12
Flexibility of schedule ($\pm$ days)	Up to 4 months prior to randomisation		- 3 months			Day 0	$\pm$ 0D	$\pm$ 1D	$\pm$ 2D	$\pm$ 4D	$\pm$ 7D	$\pm$ 7D	$\pm$ 7D	$\pm$ 14D	$\pm$ 14D	$\pm$ 14D	$\pm$ 14D
Patient Global Impression of Change and Clinician Global Impression of Change														•	•		•
Treatment Experience Interview ⁹															•		•
Visual acuity ⁴	•		•	•	•		•	•	•	•	•	•	•	•	•	•	•
Low Luminance Visual Acuity			•	•	•									•	•	•	•
Contrast sensitivity	•		•	•	•									•	•	•	•
Reading speed			•	•	•									•	•	•	•
Colour vision assessments			•	•	•									•	•		•
Static visual fields			•	•	•									•	•		•
Microperimetry			•	•	•									•	•		•
Ocular examination	•		•	•	•		•	•	•	•	•	•	•	•	•	•	•
Fundus photography			•				•	•	•	•	•	•	•	•	•	•	•
Optical coherence tomography	•		•	•	•		•	•	•	•	•	•	•	•	•	•	•
Adaptive optics imaging			•												•		•
Fundus autofluorescence			•	•	•									•	•		•
Full field electroretinography ⁵			•												•		•
Pattern electroretinography ⁵			•												•		•
Multifocal electroretinography ⁵			•												•		•
Visual mobility assessment ⁸	•													•	•		•

Table 4: Trial Assessments (Deferred Treatment arm) *

	Screening Up to 4 months prior to randomisation	Randomisation	Baseline ^{1,2}			M3 ⁸	M6 ⁸	ATIMP admin after M6 assessments	D1 post-ATIMP admin	D3 post-ATIMP admin	W1 post-ATIMP admin	W2 post-ATIMP admin	W4 post-ATIMP admin	W6 post-ATIMP admin	W9 post-ATIMP admin	M3 post-ATIMP admin	M6 post-ATIMP admin
Flexibility of schedule (± days)						± 14D	14D	+14D	± 0D	± 1D	± 2D	± 4D	± 7D	± 7D	± 7D	± 14D	± 14D
Visit number	1		2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Informed consent	•																
Physical exam	•						•										
Medical history	•					•	•				•	•	•	•		•	•
Eligibility determination	•						•										
IMP administration								★									
Adverse event review			•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Concomitant medication review	•		•	•	•	•	•	•	•	•	•	•	•	•	•	•	•
Genetic screening	•																
Vital signs (including height and weight) ³	•		•				•		•	•	•	•	•	•	•		
Randomisation		★															
Haematology			•				•		•		•	•	•	•			
Biochemistry/ glucose			•				•		•		•	•	•	•	• ⁷		
Hepatitis B (HBV) testing ⁶	•														•		
Serology			•										•			•	•
PCR									•				•				

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	Screening Up to 4 months prior to randomisation	Randomisation	Baseline ^{1,2}			M3 ⁸	M6 ⁸	ATIMP admin after M6 assessments	D1 post-ATIMP admin	D3 post-ATIMP admin	W1 post-ATIMP admin	W2 post-ATIMP admin	W4 post-ATIMP admin	W6 post-ATIMP admin	W9 post-ATIMP admin	M3 post-ATIMP admin	M6 post-ATIMP admin
Flexibility of schedule ($\pm$ days)						$\pm 14D$	14D	+14D	$\pm 0D$	$\pm 1D$	$\pm 2D$	$\pm 4D$	$\pm 7D$	$\pm 7D$	$\pm 7D$	$\pm 14D$	$\pm 14D$
QoL questionnaires (IVI, EQ5D-5L, EQ5D-Y, Low Luminance Questionnaire), Patient Global Impression of Severity			•				•									•	•
Patient Global Impression of Change and Clinician Impression of Change																•	•
Treatment Experience Interview ¹⁰																	•
Visual acuity ⁴	•		•	•	•	•	•		•	•	•	•	•	•	•	•	•
Low Luminance Visual Acuity			•	•	•											•	•
Contrast sensitivity	•		•	•	•	•	•									•	•
Reading speed			•	•	•	•	•									•	•
Colour vision assessments			•	•	•	•	•									•	•
Static visual fields			•	•	•	•	•									•	•
Microperimetry			•	•	•	•	•									•	•
Ocular examination	•		•	•	•	•	•		•	•	•	•	•	•	•	•	•
Fundus photography			•			•	•		•	•	•	•	•	•	•	•	•

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	Screening Up to 4 months prior to randomisation	Randomisation	Baseline ^{1,2}			M3 ⁸	M6 ⁸	ATIMP admin after M6 assessments	D1 post-ATIMP admin	D3 post-ATIMP admin	W1 post-ATIMP admin	W2 post-ATIMP admin	W4 post-ATIMP admin	W6 post-ATIMP admin	W9 post-ATIMP admin	M3 post-ATIMP admin	M6 post-ATIMP admin
Flexibility of schedule ($\pm$ days)						$\pm$ 14D	14D	+14D	$\pm$ 0D	$\pm$ 1D	$\pm$ 2D	$\pm$ 4D	$\pm$ 7D	$\pm$ 7D	$\pm$ 7D	$\pm$ 14D	$\pm$ 14D
Optical coherence tomography	•		•	•	•	•	•		•	•	•	•	•	•	•	•	•
Adaptive optics imaging			•				•										•
Fundus autofluorescence			•	•	•	•	•									•	•
Full field electroretinography ⁵			•			•											•
Pattern electroretinography ⁵			•			•											•
Multifocal electroretinography ⁵			•			•											•
Visual mobility assessment ^{8,9}	•					•	•									•	•

¹Baseline assessments may be performed on the same day where considered logistically and clinically appropriate. All assessments should be performed as soon as possible however all baseline assessments must be done within a 4 month period.

²Additional baseline assessments are permitted if there is inconsistency between the screening and first baseline assessment

³Measurements for vital signs include: blood pressure, pulse, respiration rate, arterial oxygen saturation, temperature, height and weight

⁴ Visual acuity at day 1 and 3 post-ATIMP administration will be assessed using the previous refraction.

⁵ERG assessments may be performed under general anaesthesia if considered appropriate

Table 5: Adverse Event Definitions

Adverse Event (AE)	Any untoward medical occurrence in a clinical trial participant administered a medicinal product and which does not necessarily have a causal relationship with this product.
Adverse Reaction (AR)	Any untoward and unintended response to an investigational medicinal product related to any dose administered.
Unexpected Adverse Reaction (UAR)	An adverse reaction, the nature or severity of which is not consistent with the applicable product information (eg Investigator's Brochure for an unauthorised product or summary of product characteristics (SPC) for an authorised product.
Serious Adverse Event (SAE) or Serious Adverse Reaction (SAR)	Any AE or AR that at any dose: • results in death • is life threatening* • requires hospitalisation or prolongs existing hospitalisation** • results in persistent or significant disability or incapacity • is a congenital anomaly or birth defect • or is another important medical condition***
SUSAR	Suspected Unexpected Serious Adverse Reaction
* the term life threatening here refers to an event in which the participant is at risk of death at the time of the event; it does not refer to an event that might hypothetically cause death if it was more severe (e.g. a silent myocardial infarction) ** Hospitalisation is defined as an in-patient admission, regardless of length of stay, even if the hospitalisation is a precautionary measure for continued observation. Hospitalisation for pre-existing conditions (including elective procedures that have not worsened) do not constitute an SAE *** Medical judgement should be exercised in deciding whether an AE or AR is serious in other situations. Important AEs or ARs that may not be immediately life threatening or result in death or hospitalisation, but may seriously jeopardise the participant by requiring intervention to prevent one of the other outcomes listed in the table (e.g. a secondary malignancy, an allergic bronchospasm requiring intensive emergency treatment, seizures or blood dyscrasias that do not require hospitalisation, or development of drug dependency)	

Adverse events include:

- an exacerbation of a pre-existing illness
- an increase in the frequency or intensity of a pre-existing episodic event or condition
- a condition (regardless of whether PRESENT prior to the start of the trial) that is DETECTED after trial drug administration. (This does not include pre-existing conditions recorded as such at baseline – as they are not detected after trial drug administration)

Clinically significant abnormalities in the results of objective tests will also be recorded as adverse events. If the results are not expected as part of disease or surgery these will also be recorded as unexpected. There are currently no expected events associated with the ATIMP.

All serious adverse events will be recorded in the hospital notes and on a paper SAE form. Adverse events will be recorded with clinical symptoms and accompanied with a simple, brief description of the event, including dates as appropriate. All adverse events will be recorded until the end of the trial (refer to Section 5.6.3 for definition), or until pregnancy outcome in the case of pregnancy if this extends beyond the end of the study. All SAEs will be recorded, fully investigated and appropriately managed until resolution or stabilisation and CI sign off.

5.11.3.4.1 Seriousness assessment

When an AE or AR occurs, the investigator responsible for the care of the participant must assess whether or not the event is serious using the definition given in Table 5: Adverse Event Definitions. If the event is classified as 'serious' then an SAE form must be completed and emailed to [REDACTED] (or delegated body) within 24 hours.

5.11.3.4.2 Severity or grading of Adverse Events

The severity of all AEs in this trial should be graded using the toxicity gradings in NIH CTCAE Version 4.0 (NIH 2009) (Table 6: Grading of Adverse Events).

Table 6: Grading of Adverse Events

Category	Definition
Mild (Grade I)	Asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
Moderate (Grade II)	Minimal, local or non-invasive intervention indicated; limiting age appropriate instrumental ADL*
Severe (Grade III)	Severe or medically significant but not immediately life threatening; hospitalisation or prolongation of hospitalisation indicated; disabling; limiting self-care ADL**
Grade IV	Life threatening consequences; urgent intervention indicated
Grade V	Death related to AE

* Instrumental ADL (Activities of Daily Living) refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.

**Self-care ADL refer to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden.

5.11.3.4.3 Causality

Causality will be assessed in terms of both the ATIMP and the surgical procedures. Based on all available information at the time of completion of the case report form, the investigator must assess the causality of all serious events or reactions. It is of particular importance in this trial to capture and differentiate events related to:

- The ATIMP administration surgery
- The ATIMP

The differentiated causality assessments will be captured in the trial specific CRF and SAE form using the definitions in Table 7: Causality Definitions.

Table 7: Causality Definitions

Relationship	Description	Event Type
Unrelated	There is no evidence of any causal relationship	Unrelated SAE
Unlikely to be related	There is little evidence to suggest that there is a causal relationship (e.g. the event did not occur within a reasonable time after administration of the trial medication). There is another reasonable explanation for the event (e.g. the participant's clinical condition or other concomitant treatment)	Unrelated SAE
Possibly related	There is some evidence to suggest a causal relationship (e.g. because the event occurs within a reasonable time after administration of the trial medication). However, the influence of other factors may have contributed to the event (e.g. the participant's clinical condition or other concomitant treatment)	SAR
Probably related	There is evidence to suggest a causal relationship and the influence of other factors is unlikely	SAR
Definitely related	There is clear evidence to suggest a causal relationship and other possible contributing factors can be ruled out.	SAR

For each AE/SAE, a causality assessment for the ATIMP administration surgery must also be provided. Any events related to the surgery alone will not be reported to the relevant regulatory agencies.

5.11.3.4.4 Expectedness

In view of the very limited clinical experience with the ATIMP there are at present no events considered as expected for the ATIMP. Therefore, any SAEs that are related to the ATIMP (i.e., considered a SAR) will be deemed a SUSAR (suspected, unexpected, serious adverse reaction) and MHRA, REC/IBC, FDA, and NIH reporting guidelines apply (see Notifications sections of the protocol).

Table 8: Assessment of expectedness

Category	Definition
Expected	An adverse event that is classed in nature as serious and which is consistent with the information about the surgery listed in the Investigator Brochure or clearly defined in the protocol. In view of the very limited clinical experience with the ATIMP there are at present no events considered as expected for the ATIMP listed in the current Investigator Brochure.
Unexpected	An adverse event that is classed in nature as serious and which is not consistent with the information about the ATIMP and surgery listed in the Investigator Brochure* or clearly defined in the protocol.

*This includes listed events that are more frequently reported or more severe than previously reported

The reference document to be used to assess expectedness against the ATIMP and surgery is the Investigator Brochure. Procedure-related adverse events cannot be considered attributable to the ATIMP.

Previous experience with AAV-mediated gene therapy in the retina indicates that the risks are largely limited to the eye. A temporary and/or mild decrease in visual acuity, due to detachment of the retina or post-surgical inflammation, is to be expected after intraocular surgery and will not cause undue discomfort to the participants. Therefore, we have defined the success criteria for the primary outcome for safety as the absence of an adverse event that has a substantial and sustained negative impact on vision, as well as the absence of any non-ocular SUSAR.

Expected events associated with surgery:

- Temporary and/or mild decrease in visual acuity (to hand movements or better for a period of up to 8 weeks), due to detachment of the retina or post-surgical inflammation
- Ocular discomfort

necessary, sign and then email to [REDACTED]. Systems will be in place at the site to enable the investigator to check the form for clinical accuracy as soon as possible.

The minimum criteria required for reporting an SAE are the trial number and date of birth, name of reporting investigator and sufficient information on the event to confirm seriousness. Any further information regarding the event that is unavailable at the time of the first report should be sent as soon as it becomes available.

The SAE form must be scanned and sent by email to the trial team on [REDACTED]

Participants must be followed up until clinical recovery is complete and laboratory results have returned to normal or baseline values, or until the event has stabilised. Follow-up should continue after completion of trial follow-up if necessary. Follow-up SAE forms (clearly marked as follow-up) should be completed and emailed to [REDACTED] as further information becomes available. Additional information and/or copies of test results etc may be provided separately. The participant must be identified by trial number and date of birth only. The participant's name must not be used on any correspondence and must be blacked out and replaced with trial identifiers on any test results should they be provided.

5.11.3.5.2 [REDACTED] responsibilities

[REDACTED] will follow [REDACTED] Standard Operating Procedures and a study specific Safety Management Plan to ensure that case processing of events occurs within appropriate regulatory timeframes. [REDACTED] will submit Development Safety Update Reports (DSURs) to regulatory authorities.

5.11.3.5.4 Reporting SUSARs in International Trials

The mechanism for reporting SUSARs that occur outside of the UK to the MHRA, and those that occur outside of the US to the FDA will be covered in the trial specific Safety Management Plan.

5.11.3.5.5 Annual Progress Reports

An annual progress report (APR) will be submitted to the UK REC within 30 days of the anniversary date on which the favourable opinion was given, and annually until the trial is declared ended. Annual IRB applications for continuing review will be submitted with sufficient time to allow review and approval of trial continuation.

8 Appendix 1: Guidance on Study Conduct during the COVID-19 Pandemic

The measures outlined in this Appendix are temporary, while access to sites is restricted during the COVID 19 pandemic. As restrictions are lifted, the decision to revert back to the protocol in effect prior to the pandemic should be discussed and agreed with the sponsor.

It is recognized that the Coronavirus Disease 2019 (COVID-19) pandemic may have an impact on the conduct of this clinical study due to, for example, self-isolation/quarantine by participants and study-site personnel; travel restrictions/limited access to public places, including hospitals; study site personnel being reassigned to critical tasks.

In alignment with recent health authority guidance, the sponsor is providing options for study related participant management in the event of disruption to the conduct of the study. This guidance does not supersede any local or government requirements or the clinical judgement of the investigator to protect the health and well-being of participants and site staff.

Every effort should be made to adhere to protocol-specified assessments for participants on study intervention, including follow up. Modifications to protocol-required assessments may be permitted after consultation between the participant and investigator, and with the agreement of the sponsor. Missed assessments/visits will be captured in the EDC for protocol deviations. Discontinuations of study interventions and withdrawal from the study should be documented with the prefix “COVID-19-related” in the case report form (CRF).

Scheduled visits that cannot be conducted in person at the study site will be performed remotely/virtually, and site -based evaluations may be delayed until such time that on-site visits can be resumed. At each contact, participants will be interviewed to collect safety data. Key efficacy and safety endpoint assessments should be performed as feasible. Participants will also be questioned regarding general health status to fulfill any physical examination requirement.

The sponsor will continue to monitor the conduct and progress of the clinical study and any changes (eg, delay or discontinuation in recruitment, site monitoring and audits) will be communicated to the sites and health authorities according to local guidance. If a participant has tested positive for COVID 19, the investigator should contact the sponsor's responsible medical officer to discuss plans for study intervention and follow-up. Modifications made to the study conduct as a result of the COVID-19 pandemic should be summarized in the clinical study report.

GUIDANCE SPECIFIC TO THIS PROTOCOL

Each patient will be risk assessed on a case by case basis by the PI and his/her research team to decide the appropriate care for the patient.

Item no	Item	Management of Item
1	Missed assessments	Missed assessments will be captured in the EDC system as a deviation with the prefix COVID 19 RELATED.
2	Protocol Deviations	Any deviations to the study protocol due to COVID 19 pandemic will be captured in the EDC system will be entered with the prefix COVID 19 RELATED.
3	Patient missing any scheduled visits	If scheduled visits cannot be conducted within the visit window the site can schedule an additional visit at the earliest opportunity the patient is able to travel,
4	Baseline Assessments	Baseline assessments for those patients who have had surgery postponed due to COVID 19 - these baseline assessments will be valid up to 6 months expect for Haematology, Biochemistry and Glucose which will need repeating.
5	Patients may be unable to attend study visit during visit window.	It is acknowledged that this is a global crisis and patients may be put at more risk attending hospital appointments due to route of travel into research facility. If appropriate the sponsor will offer expenses for taxi fares to avoid busy train or bus routes. If patients are unable to reach their research facility but are able to seek help at a local ophthalmologist for urgent concerns this is acceptable, and the sponsor will encourage patients to show any non-trial staff conducting assessments the contact details for the trial staff so that conversations can be had between the trial and non-trial medical staff
6	Patients visits conducted outside of visit window	Patients will be assessed individually by the sites on a case by case basis and phone interviews may be conducted by sites for all follow up visits up until at least 3 months post-surgery.

		Patients undergoing telephone assessment will be asked to provide information on any adverse events or other safety concerns.
7	Patients self-quarantined or government implemented quarantine.	If patients and/or their carer are under quarantine and unable to leave home, their cases will be assessed by medical monitors and PI as to risk of delayed appointment. (e.g ongoing AEs, date of last appointment, less than 3 months since intervention).
8	Potential lack of reporting of symptoms due to delayed visits Delayed reporting of SAEs, SARs & SUSARs- Potential for events to not be reported within 24 hours of occurrence Delayed Participant reporting of Adverse Events	All participants are provided with emergency cards on enrolment to the study to enable 24-hour contact with study investigators. All events should be reported by an Investigator (as assigned on the delegation log) or another member of the site team, in the Investigators absence and where delegated, within 24 hours of the event occurring. Reminder of importance to report any new AEs/symptoms to be communicated to patients by the sites during this time.
9	Emergencies during the study and the provision of emergency contact arrangements:	If the participants facing difficulties in contacting the Trial Team in the event of emergencies, they have been provided with an alert card to inform other healthcare providers of their participation in the gene therapy trial. The Trial card includes the 24/7-hour contact number, patients trial code, trial number, brief details of the trial intervention and EudraCT number.
10	Patients may require NIMPs that they would have collected a scheduled study visit.	Site research staff will be in regular contact with their study patients and should they require NIMPs, the site pharmacy will be able to courier them to the patient and charge the cost to the sponsor.
11	Assessment results outside of window not true representation of protocol stated timepoints	Patient safety will be most important factor to consider. Patients will be invited as close to visit window as possible and deviations recorded where applicable.
12	Questionnaire responses could be incorrect due to expected visit vs actual visit attended.	Patient may confuse true answers if questionnaire asks for certain timepoints. Site research staff will explain questionnaire details before patient answers.
13	Clinician/s and site research staff unavailable due to contraction of virus	Site to ensure appropriate staff are available and documented on the delegation log to provide medical cover if the PI is unavailable.
14	Monitor may not be allowed on site or may be quarantined and unable to visit.	Monitors will be encouraged to attend site where possible, if unable will conduct remote monitoring as much as possible and arrange as many visits as is required to get back on track once site re-opens.

9 Protocol Amendments

Protocol Version and Date	Reason for Amendment
Protocol Version 1	NA
Protocol Version 2 10 March 2017	To incorporate the flow chart for the dose escalation, dose of drugs used at the time of administration of the ATIMP and reconsenting the children as they progress through the age brackets.
Protocol Version 3 12 April 2017	To extend the course of post-surgery prophylactic steroids from 4 weeks to 8 weeks. Consequently, the duration for considering dose limiting events has been extended from 6 weeks to 9 weeks to cover the period of steroid administration and one additional week and other minor clarifications
Protocol Version 4 25 May 2017	To add additional 2 exclusion criteria, and to refer to a barrier and spermicide form of contraception, rather than double barrier method and also to clarify that only males will be included in the trial
Protocol Version 5 13 November 2017	To update the medium and high dose. To reduce the gap between participant 1 and 2/3 in a cohort from 9 weeks to 4 weeks and update the prophylactic post-administrative steroid regimen in children, and clarify safety reporting and confirmatory safety dose for children
Protocol Version 6 27 February 2018	To clarify the allowance of data obtained from the natural history study be used for screening and or baseline assessments (with consent from subjects) in order to avoid unnecessary testing of subjects. Clarify that more than 1 surgeon at a site may inject vector
Protocol Version 7.0 26 April 2018	To clarify number of visual mobility assessments required To reduce the specificity of anti-biotics that may be prescribed on the trial.
Protocol Version 7.1 19 September 2018 (MEEI Specific)	This amendment was specific to Massachusetts Eye and Ear Institute to include the need for Hepatitis B screening.
Protocol Version 9.0 11 February 2019	To include a randomised component to the expansion phase of the study To include further QOL measurement tool To reduce the follow up from 18 to 12 months

	To include an Interim Analysis at 3 and 6 months post treatment for patients in the low/intermediate dose treatment arms of the randomised component of the trial
Protocol Version 10.0 09 August 2019	To amend and clarify the inclusion and exclusion criteria for the trial To remove FST at all assessments and to remove ERG at screening to reduce the assessment burden on patients To add in Low Luminance Visual Acuity testing To include additional Patient Reported Outcomes and to add a Clinician Reported Outcome. To include Treatment Experience interviews. To correct minor errors in the Patient Visit Schedule
Protocol v11.0 01 May 2020	Included guidance on the management of patients during COVID 19 pandemic Corrected minor errors in the protocol v10.0 Updated Steroids risks Included update local steroid use